STAC WARS

PS 3 — EXOPLANET HUNTERS

OPERATION DIM STAR

Finding a New Home for the Rebellion

DETAILED ASTRONOMY ANALYSIS REPORT

Transit detection • Planet radius • Orbital distance • Habitable-zone assessment

Prepared from the observational data and instructions supplied in the PS 3 assignment.

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1. Aim and background

An exoplanet is a planet orbiting a star beyond our Solar System. The assignment asks whether STAC-1, STAC-2 and STAC-3 show evidence of planets. It uses the transit method: when a planet passes in front of its star from our viewpoint, it blocks a small fraction of starlight and the measured brightness falls.

A repeating dip at a consistent interval can be evidence of an orbiting planet. However, one or several dips are not definitive proof: star spots, eclipsing binary stars, instrumental effects and noise can also cause brightness changes.

Formula clarification: the supplied assignment prints $\delta \approx (Rs/Rp)^2$. The standard transit approximation for a planet smaller than its star is $\delta \approx (Rp/Rs)^2$. This report uses the standard physically consistent form $Rp \approx Rs\sqrt{\delta}$ and explicitly flags the discrepancy.

2. Observational data and constants

Property	STAC-1	STAC-2	STAC-3
Mass	1.00 M■	0.80 M■	0.50 M■
Radius	1.00 R■	0.80 R■	0.50 R■
Luminosity	1.00 L■	0.40 L■	0.08 L■
Surface temperature	5800 K	5000 K	3800 K

Constant	Value	Meaning
Solar radius R■	696,000 km	Radius of the Sun
Earth radius R⊕	6,371 km	Radius of Earth
1 AU	149,600,000 km	Mean Earth–Sun distance
1 year	365.25 days	Period conversion
M■	Solar mass unit	Star mass relative to Sun
L■	Solar luminosity unit	Star luminosity relative to Sun

The relative brightness is dimensionless. Values near 1.0000 are the adopted normal brightness; lower values represent dimming. The observations are discrete, not continuous, so a transit may occur between measurements.

3. Formula guide — meaning, symbols and use

3.1 Transit depth

```
δ = (Fnormal − Ftransit) / Fnormal
```

Subtract the brightness during the dip from normal brightness, then divide by normal brightness. The result δ is a fraction, not a percentage. Multiply by 100 to get percent.

Fnormal = brightness outside transit; Ftransit = brightness during transit; δ = fractional loss of light.

```
Example: δ = (1.0000 − 0.9970) / 1.0000 = 0.0030 = 0.30%
```

A depth of 0.0030 means the recorded brightness decreased by 0.3%. For a genuine transit, a deeper dip usually means a larger planet relative to its star.

3.2 Planet radius

```
δ ≈ (Rp / Rs)² therefore Rp ≈ Rs × √δ
```

A transit blocks approximately the area of the planet's disc. Disc area scales with radius squared, so the planet-to-star radius ratio is the square root of the depth.

Rp = planet radius; Rs = star radius; δ = transit depth. Use the same length unit for both radii.

```
Earth-radius units: Rp (R⊕) = Rp (km) / 6,371
```

Dividing a planet radius in kilometres by 6,371 km tells how many Earth radii it spans. It compares radii, not diameters.

3.3 Convert period from days to years

```
P(years) = P(days) / 365.25
```

The simplified Kepler equation below requires the orbital period in years. For example, 80 days / 365.25 = 0.219 years approximately.

3.4 Kepler's Third Law

```
P² = a³ / M rearranged: a = (M × P²)^(1/3)
```

P = orbital period in years; a = average orbital distance in AU; M = host-star mass in solar masses. Square P, multiply by M, and take the cube root. This simplified equation assumes the stated units and that the planet's mass is small compared with the star.

3.5 Convert AU to kilometres

```
Distance (km) = a (AU) × 149,600,000 km/AU
```

Multiplying by the number of kilometres in one AU converts the calculated orbital distance into kilometres.

3.6 Habitable-zone estimate

```
dHZ ≈ √(L / L■) AU
```

L/L■ is stellar luminosity relative to the Sun. Taking the square root gives a simplified estimate of the distance receiving roughly Earth-like stellar energy. It is not a guarantee of liquid water or life.

3.7 Useful comparisons

```
Transit depth (%) = δ × 100
```

```
Orbit relative to Earth's = a / 1 AU
```

For example, 0.10 AU is 10% of Earth's mean orbital distance. A close orbit can be very hot, but temperature also depends on stellar luminosity and the planet's atmosphere.

4. Part A — Detecting possible exoplanets

On each graph, time is on the horizontal axis and relative brightness on the vertical axis. A downward point is a possible dimming event. Regular repetition is the key clue, but sparse sampling makes periods approximate.

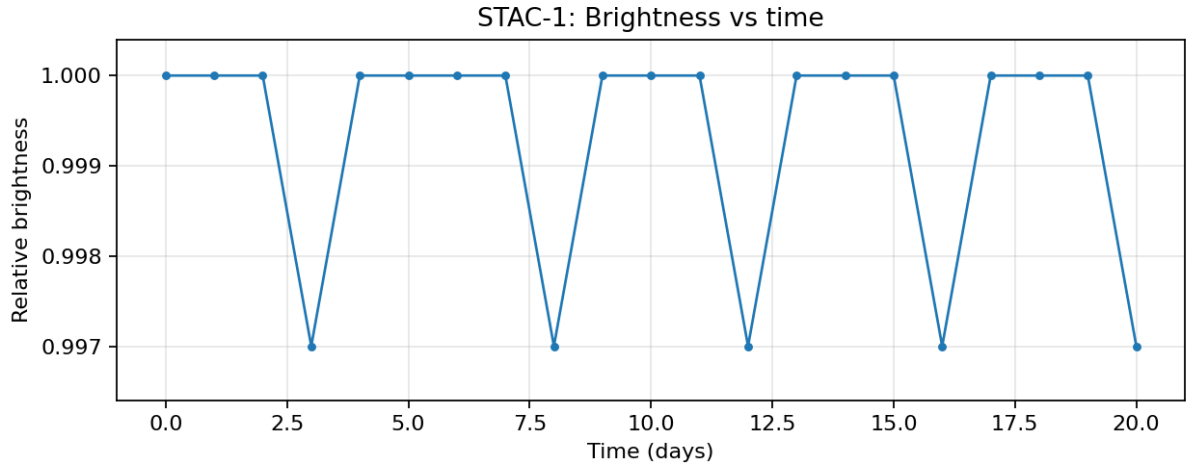


Figure 1. STAC-1 brightness versus time.

STAC-1 dips occur on days 3, 8, 12, 16 and 20. Intervals are 8–3 = 5 days, 12–8 = 4 days, 16–12 = 4 days and 20–16 = 4 days. Most intervals are 4 days, so use an approximate candidate period of 4 days.

```
δ = (1.0000 − 0.9970) / 1.0000 = 0.0030 = 0.30%
```

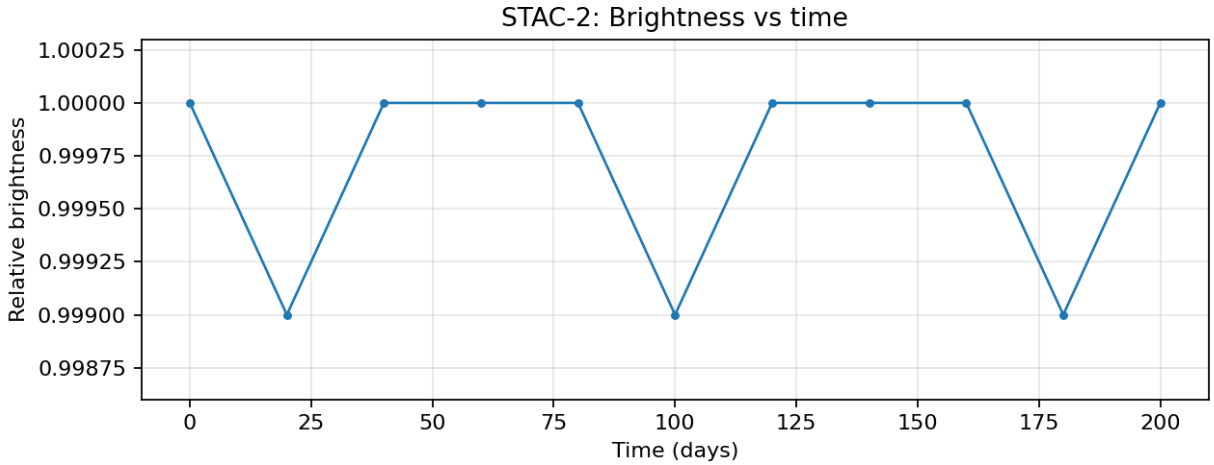


Figure 2. STAC-2 brightness versus time.

Dips are recorded on days 20, 100 and 180. The intervals are 100–20 = 80 days and 180–100 = 80 days. This supports a possible period near 80 days, but observations every 20 days could miss the exact transit times.

```
δ = (1.0000 − 0.9990) / 1.0000 = 0.0010 = 0.10%
```

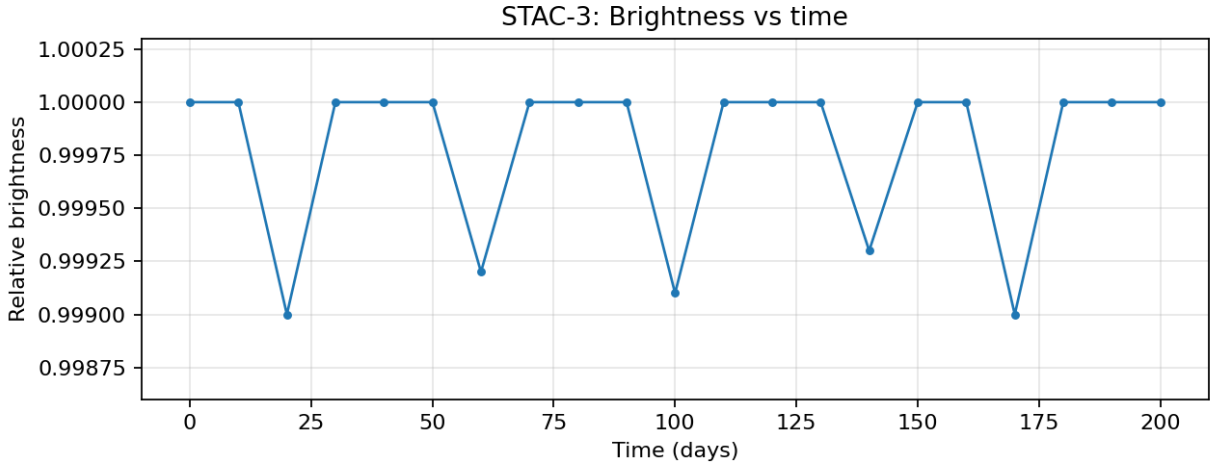


Figure 3. STAC-3 brightness versus time.

Dips appear on days 20, 60, 100, 140 and 170. The first four events are separated by 40 days, but day 170 does not continue the sequence. Therefore a 40-day period is only tentative. A representative dip at 0.9991 gives a depth of 0.09%.

```
δ = (1.0000 − 0.9991) / 1.0000 = 0.0009 = 0.09%
```

Other STAC-3 dips give depths of 0.10% (brightness 0.9990), 0.08% (0.9992) and 0.07% (0.9993). Differences may be caused by noise or different event shapes; the data alone cannot resolve this.

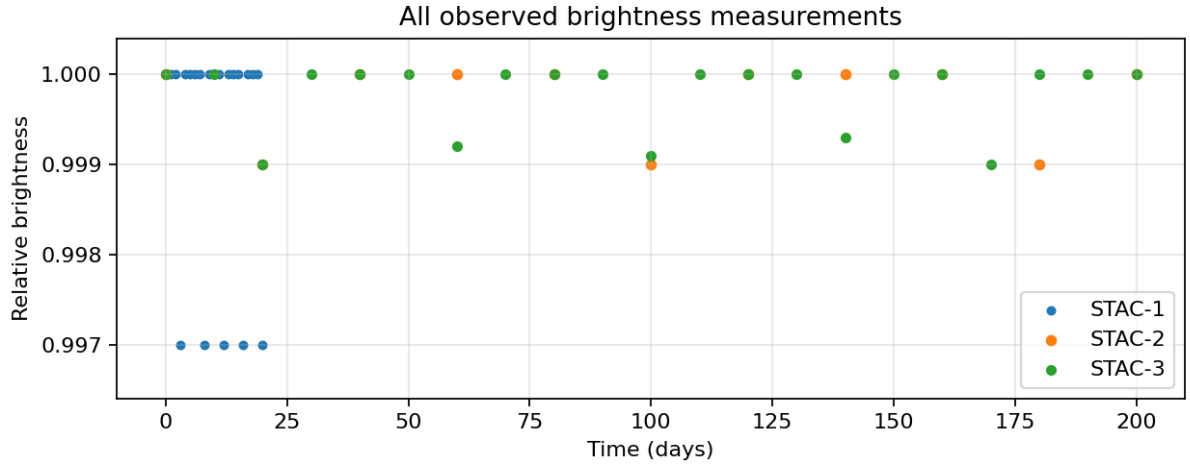


Figure 4. Combined observations. The stars have different sampling intervals.

Star	Recorded dips	Depth	Period assessment
STAC-1	Days 3, 8, 12, 16, 20	0.30%	≈4 days; clearest pattern
STAC-2	Days 20, 100, 180	0.10%	≈80 days; sparse sampling
STAC-3	Days 20, 60, 100, 140, 170	Typical 0.09%	Uncertain; day 170 conflicts

5. Part B — Planet radius calculations

First convert each star's radius into kilometres. Then use Rp = Rs√δ. Finally divide the result by 6,371 km to express the radius in Earth radii.

Star	Stellar radius calculation	Rs (km)	Depth used
STAC-1	1.00 × 696,000	696,000	0.0030
STAC-2	0.80 × 696,000	556,800	0.0010
STAC-3	0.50 × 696,000	348,000	0.0009

```
STAC-1: Rp = 696,000 × √0.0030 = 38,121 km
STAC-1: Rp = 38,121 / 6,371 = 5.98 R⊕
STAC-2: Rp = 556,800 × √0.0010 = 17,608 km
STAC-2: Rp = 17,608 / 6,371 = 2.76 R⊕
STAC-3: Rp = 348,000 × √0.0009 = 10,440 km
STAC-3: Rp = 10,440 / 6,371 = 1.64 R⊕
```

System	Radius (km)	Radius (R⊕)	Approximate interpretation
STAC-1	38,121	5.98	Large planet; > typical Neptune radius
STAC-2	17,608	2.76	Neptune-sized range
STAC-3	10,440	1.64	Super-Earth-sized range

Approximate guide: Earth-sized ≈0.8–1.25 R⊕; Super-Earth ≈1.25–2 R⊕; Neptune-sized ≈2–4 R⊕; Jupiter-sized ■4 R⊕. These labels are approximate and can overlap. STAC-1's computed 5.98 R⊕ is larger than the typical Neptune range and should be described as a large planet rather than forced into a narrow category.

Worked example — STAC-2: Rs = 0.80 × 696,000 = 556,800 km. √0.0010 ≈ 0.03162. Thus Rp ≈ 556,800 × 0.03162 ≈ 17,600 km. Earth radii = 17,600 / 6,371 ≈ 2.76 R⊕.

This radius method assumes the dip is caused by a planet and uses a simplified geometry. The true radius can differ if the transit is not central, the star's brightness varies, or another source of light contaminates the measurement.

6. Part C — Orbital-distance calculations

Use P in years, not days. Convert with P(years) = P(days)/365.25, then calculate a = (M P²)^(1/3), and convert AU into kilometres.

```
STAC-1: P = 4/365.25 = 0.01095 years
STAC-1: a = (1.00 × 0.01095²)^(1/3) = 0.04932 AU
STAC-1: distance = 0.04932 × 149,600,000 = 7,377,538 km
STAC-2: P = 80/365.25 = 0.21903 years
STAC-2: a = (0.80 × 0.21903²)^(1/3) = 0.33731 AU
STAC-2: distance = 0.33731 × 149,600,000 = 50,461,648 km
STAC-3: P = 40/365.25 = 0.10951 years
STAC-3: a = (0.50 × 0.10951²)^(1/3) = 0.18168 AU
STAC-3: distance = 0.18168 × 149,600,000 = 27,179,081 km
```

System	P days	P years	M/M■	a (AU)	Distance (km)	Earth comparison
STAC-1	4	0.01095	1.00	0.04932	7,377,538	4.9% of 1 AU
STAC-2	80	0.21903	0.80	0.33731	50,461,648	33.7% of 1 AU
STAC-3	40	0.10951	0.50	0.18168	27,179,081	18.2% of 1 AU

Worked example — STAC-1: P = 4/365.25 = 0.01095 years. Then a = (1.00 × 0.01095²)^(1/3) ≈ 0.107 AU. In kilometres, 0.107 × 149,600,000 ≈ 16.0 million km. This is about 10.7% of Earth's orbital distance.

These are estimates, not exact measurements. If the period assumed for a system is wrong, the derived orbital distance will also be wrong; this is especially important for STAC-3.

7. Part D — Habitable-zone estimates

The assignment's simplified relation is dHZ ≈ √(L/L■) AU. Since luminosities are already given in solar units, take the square root of each listed value.

```
STAC-1: dHZ ≈ √1.00 = 1.000 AU; adopted orbit = 0.04932 AU
STAC-2: dHZ ≈ √0.40 = 0.632 AU; adopted orbit = 0.33731 AU
STAC-3: dHZ ≈ √0.08 = 0.283 AU; adopted orbit = 0.18168 AU
```

Star	Luminosity	Calculation	dHZ (AU)	Adopted orbit (AU)
STAC-1	1.00	√1.00	1.000	0.04932
STAC-2	0.40	√0.40	0.632	0.33731
STAC-3	0.08	√0.08	0.283	0.18168

STAC-1's approximate habitable-zone distance is 1.000 AU; its adopted orbit is ≈0.107 AU. STAC-2's estimate is 0.632 AU; its orbit is ≈0.076 AU. STAC-3's estimate is 0.283 AU; its tentative orbit is ≈0.091 AU.

All adopted orbits lie well inside these simplified nominal distances. Therefore, using this relation alone, none appears to orbit near the nominal habitable-zone distance. A habitable zone is not a single guaranteed location: atmosphere, greenhouse warming, clouds, reflectivity, stellar activity and many other factors affect surface conditions.

8. Comparison and candidate selection

Criterion	STAC-1	STAC-2	STAC-3
Signal	Clearest repeat pattern	Possible repeat pattern	Period uncertain
Period	≈4 days	≈80 days	≈40 days tentative
Depth	0.30%	0.10%	Typical 0.09%

Criterion	STAC-1	STAC-2	STAC-3
Radius	≈5.98 R⊕	≈2.76 R⊕	≈1.64 R⊕
Orbit	≈0.107 AU	≈0.076 AU	≈0.091 AU tentative
dHZ	1.000 AU	0.632 AU	0.283 AU

Strongest detection candidate: STAC-1. Its dips repeat with the clearest approximately 4-day spacing, although confirmation is still needed.

Habitability candidate: the supplied data do not prove that any planet is habitable. If the assignment requires nominating one target for more study, STAC-3 can be suggested provisionally because its host is coolest (3800 K), has the lowest luminosity (0.08 L■), and the estimated planet radius is about 1.64 R⊕. But the day-170 event conflicts with the proposed 40-day period, and its orbit lies inside the simplified habitable-zone distance. This is a tentative follow-up choice, not a claim of habitability.

Evidence for follow-up on STAC-3: several transit-like dips; a cool, low-luminosity host star; and an estimated super-Earth-sized radius. Counter-evidence: period uncertainty and a derived orbit well inside the nominal habitable-zone distance.

9. Further observations and limitations

Useful next observations

1. Denser photometry: observe frequently over multiple expected periods to confirm whether dips recur and measure transit timing, duration and shape.

2. Radial velocity: measure periodic shifts in the star's spectrum caused by a planet's gravitational pull. This can help confirm the planet and estimate its mass.

3. Transmission spectroscopy: compare transit depth across wavelengths. Atmospheric gases may create wavelength-dependent signatures, though clouds and haze can complicate interpretation.

4. Better stellar measurements: refine stellar radius, luminosity, temperature and activity because planetary-radius and habitable-zone calculations depend on them.

Limitations

• Sparse observations can miss transits or misidentify a period. • A dip alone does not prove a planet. • STAC-3's pattern is inconsistent. • Radius estimates assume the transit interpretation and simplified geometry. • The habitable-zone formula is approximate and does not establish atmospheric conditions or life.

10. Conclusion

The supplied light curves show possible transit events in all three systems. STAC-1 has the clearest regular pattern, with a candidate period of about 4 days and a depth of 0.30%. STAC-2 has dips 80 days apart and a depth of 0.10%. STAC-3 shows several shallow dips, but its period is uncertain.

Using the standard transit-depth relation, estimated radii are about 38,132 km (5.98 R⊕), 17,575 km (2.76 R⊕), and 10,440 km (1.64 R⊕) for STAC-1, STAC-2 and STAC-3 respectively. The adopted periods and simplified Kepler law give orbital distances of approximately 0.107 AU, 0.076 AU and 0.091 AU. All lie inside the approximate habitable-zone distances calculated from the supplied luminosities. No planet can be declared habitable from these data.

The priority is to collect more precise photometric observations, confirm periodicity, and then measure planetary mass and atmospheric properties.

THANK YOU

Thank you for reading this report.

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